

A High-Sensitivity Surge-Mode RF Energy Harvester for Self-Powered Devices

Xing Liu^{1,2}, Hui Wu^{1,2}, Jinbo Chen^{1,2}, Wenjun Zou^{1,2},
Yijun Ye^{1,2}, Jie Yang^{1,2}, and Mohamad Sawan^{1,2}, *Life Fellow, IEEE*

¹ CenBRAIN Neurotech, School of Engineering, Westlake University, Hangzhou, Zhejiang, China

² Integrated-On-Chips Brain-Computer Interfaces Zhejiang Engineering Research Center, Hangzhou, Zhejiang, China

Abstract—We present in this paper a high-sensitivity surge-mode radio-frequency energy harvesting front-end for self-powered devices. The proposed architecture integrates an eight-stage cross-coupled differential-drive rectifier, a nanowatt-level voltage monitor, and an ultra-low-power power management unit featuring a CMOS voltage reference and adaptive flipped-voltage-follower LDO. By enabling intermittent surge-mode operation, the system effectively transforms the rectifier’s load-dependent behavior into an ultra-light-load condition, achieving efficient energy accumulation under weak RF inputs. Implemented in a 40-nm CMOS process, the proposed front-end achieves a sensitivity of -22 dBm and a peak power conversion efficiency of 42.25% at 915 MHz, providing a stable and adaptive power supply for long-term maintenance-free IoT and biomedical applications.

Index Terms—RF energy harvesting, surge-mode operation, low-power PMU, adaptive LDO, low-power techniques.

I. INTRODUCTION

The proliferation of the Internet of Things (IoT) and the rapid development of implantable and wearable medical devices have driven the demand for large-scale, battery-less, and self-powered sensor networks capable of operating maintenance-free for decades [1]. Radio-frequency energy harvesting (RFEH) has emerged as a promising enabling technology owing to its ubiquity and full compatibility with standard CMOS integration [2]. However, the extremely weak and fluctuating nature of ambient RF power leads to unstable output and limited driving capability in corresponding harvesting systems, making it difficult for conventional always-on circuits to operate reliably.

To address these challenges, previous studies have proposed various compensation techniques that provide auxiliary gate biasing in cross-coupled differential-drive (CCDD) rectifiers to overcome the threshold voltage limitation, thereby enhancing sensitivity and efficiency [3], [4]. Others have focused on rectifier reconfigurations to achieve a wider high-efficiency range and stable output [5], [6]. Nonetheless, under realistic ultra-low-power (ULP) conditions, these systems still face insufficient instantaneous drive capability, failing to deliver adequate power for sensor nodes with communication demands. Inspired by low-power circuit design principles, wake-up architectures have emerged as a practical solution for future self-powered systems [7]. In such systems, the circuit alternates between sleep and active modes to conserve energy. Through intermittent operation, the effective load periodically switches between light and heavy states. As illustrated in Fig.

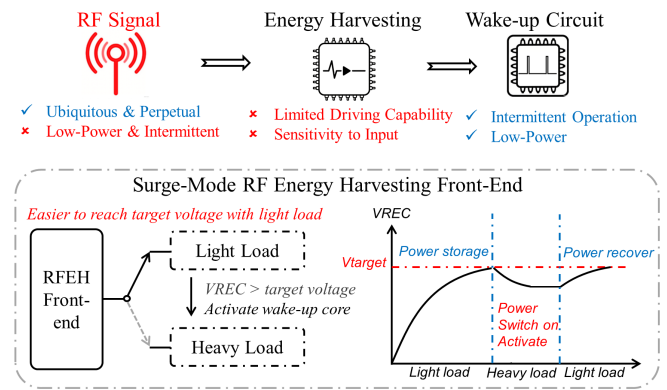


Fig. 1. Principle of surge-mode RF energy harvesting front-end for non-stop self-powered systems.

1, the RFEH front-end accumulates energy efficiently under light-load conditions until the output voltage reaches a target level, subsequently activating the functional circuits to perform predefined tasks. After task completion, the functional circuits enter a sleep state, awaiting the next activation. The duration of this dormant period varies with input conditions, yielding a wave-like operational profile that is periodic yet continuous, a state referred to as surge mode.

In recent years, several works have attempted to implement similar concepts. For instance, [8] demonstrated a design achieving -36.5 dBm sensitivity using a 108-stage Dickson rectifier combined with an on/off power management technique; however, the excessive number of stages resulted in low efficiency. In another study [9], a burst-charging RFEH system was introduced, in which a low-frequency charge pump (LFCP) with MPPT enhanced the rectifier’s output voltage to 1 V. The burst-charging strategy periodically enabled and disabled the LFCP to alleviate its load impact on the rectifier, achieving -31 dBm sensitivity, though the design remained complex and with a limited input range.

To overcome these limitations, we proposed a high-sensitivity surge-mode self-powered system based on RFEH in this paper. By integrating ULP monitor, power management unit (PMU), and a high-efficiency rectifier, the proposed architecture achieves superior sensitivity while extending the operational range of self-powered systems. The design transforms the traditional “drivability” constraint into a frequency-adaptive dynamic behavior, enabling the system to operate

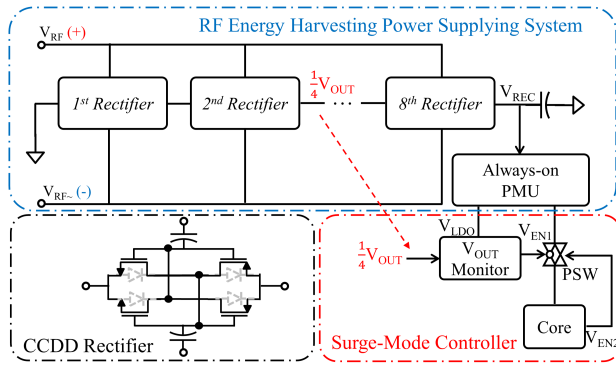


Fig. 2. System blocks of proposed surge-mode RF energy harvesting front-end and single-stage CCDD rectifier applied in system.

adaptively in surge mode according to input variations. This approach provides a stable and efficient power supply for long-term operation of low-power sensor nodes across diverse application scenarios.

The remainder of this paper is organized as follows. Section II introduces the proposed system architecture and circuit implementation, Section III presents experimental results, and Section IV concludes the paper.

II. PROPOSED RFEH FRONT-END

Fig. 2 illustrates the proposed surge-mode RFEH power supplying system. The high-efficiency RF rectifier converts the incident RF signal into DC power and charges a large off-chip storage capacitor, which in turn powers an ULP voltage reference and an adaptive LDO regulator that remain continuously active. During the charging phase, the LDO is isolated from the load through a power switch (PSW), operating in an ULP idle mode to minimize static consumption. In addition, a voltage monitoring circuit is connected to the second-stage output node of the rectifier, where a reduced supply voltage both lowers its power consumption and enables real-time monitoring of the capacitor's charging status. Once the capacitor voltage reaches a predefined threshold, the monitoring circuit triggers the PSW, connecting the LDO to the load and activating the functional circuitry. In the present prototype, since no specific functional block is integrated, a low threshold level is set to automatically deactivate the PSW after a short interval, initiating a new charging-activation cycle. In practical applications, however, the functional circuit itself can issue a termination signal to switch off the PSW and re-enter the charging phase autonomously.

Through this level-triggered, wave-like RF energy harvesting and powering scheme, the proposed architecture transforms the conventional rectifier’s load-dependent sensitivity constraint into an ultra-light load condition, consisting only of an nW-level PMU and a pW-level monitor load. As a result, the RF energy harvester can reach its target activation voltage even under extremely low input power conditions.

A. Cross-Coupled Differential Drive Rectifier Design

Fig. 2 (left-bottom) shows the conventional fixed-stage CCDD rectifier adopted in the proposed system. Benefiting

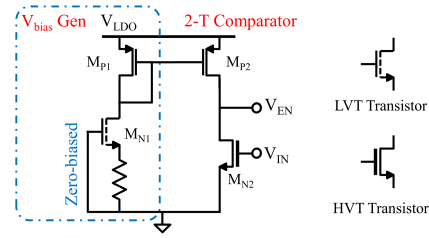


Fig. 3. Vout monitor in proposed system.

from the dynamic self-compensation inherent in the cross-coupled structure, the rectifier achieves high sensitivity and conversion efficiency under low input power conditions without requiring any additional biasing or compensation circuitry. The rectifier employs an eight-stage cascaded configuration, which offers an optimal trade-off between charging speed and power conversion efficiency (PCE).

It is noteworthy that, due to the presence of the trigger voltage in the proposed surge-mode RFEH system, the rectifier's maximum output voltage is inherently limited. Consequently, even under high RF input power conditions, the rectifier output remains stabilized around the trigger voltage. This design not only enhances system stability but also suppresses leakage currents that typically degrade the performance of fixed-stage rectifiers at high input levels, thereby improving power transfer efficiency in strong RF environments. Depending on the available input energy, the system can adaptively activate or deactivate the subsequent circuitry to maintain efficient operation.

B. Vout Monitor Design

To achieve fast charging and timely response, the voltage monitoring module must exhibit both low power consumption and high-speed operation. Therefore, simply minimizing the static current is insufficient for optimal performance. Benefiting from the identical output voltage of each stage in the multi-stage rectifier, the conventional voltage detection, which is typically performed at the rectifier output node, is shifted to the output node of the second stage. The voltage at this node is one-fourth of the total 8-stage rectifier output voltage, thereby reducing the supply voltage for the detection circuit to save energy while maintaining normal system operation.

In this system, the operating voltage is set to 1.6 V; therefore, the target detection voltage at the second-stage output node of the rectifier is 0.4 V. A number of low-power voltage detectors have been proposed in prior works [9], [10], most of which rely on CMOS 2-transistor (2T) voltage reference architectures to generate an appropriate reference voltage under low-supply conditions. Although such detectors can achieve ultra-low quiescent power, the need for level shifters often introduces additional power overhead.

In the proposed design, this issue is eliminated by integrating the voltage monitoring function within the on-chip PMU. As shown in Fig. 3, the second-stage rectifier output is directly used as the input of an nW-level comparator, which generates the activation and deactivation signals to control

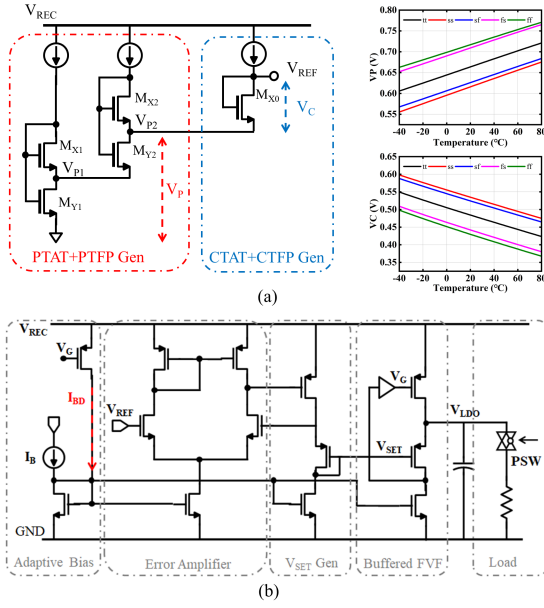


Fig. 4. (a) The proposed CMOS voltage reference (VR) and temperature characteristic of V_P and V_C across the five process corners. (b) The proposed adaptive flipped voltage follower (FVF) low-dropout (LDO) regulator.

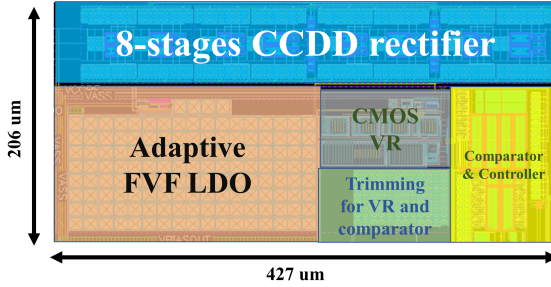


Fig. 5. Layout of proposed surge-mode RFEH front-end.

the system operation without requiring extra biasing or level-shifting circuitry.

C. Power Management Unit

To accelerate the operating frequency of the surge-mode RFEH system, the PMU, as one of the major loads, must exhibit an extremely low static current. However, low quiescent current typically comes at the expense of slower transient response. In the proposed surge-type RFEH power supplying system, the load circuits are frequently switched between active and sleep modes. If the low-power PMU and functional circuits were repeatedly turned on and off together, the slow settling time would cause extra energy and time losses.

To ensure fast and stable system recovery upon activation, the proposed system keeps the PMU continuously active, providing the necessary reference and supply voltages. Only a single power switch is used to isolate the LDO and the functional circuits that require activation.

The PMU consists of a process-compensated CMOS voltage reference and an adaptively biased flipped voltage follower (FVF) LDO, designed to achieve ultra-low static current, fast

transient response, and stable output voltage [11]–[14]. The simplified circuit diagrams are shown in Fig. 4 (a) and Fig. 4 (b). In the CMOS voltage reference, all transistors operate in the subthreshold region due to the extremely small bias current. So, we can get the following expressions for V_{P1} and V_C by the subthreshold current equations:

$$V_{P1} = nV_T \ln \left[\frac{N_{X1}W_{X1}L_{Y1}I_{Y1}}{N_{Y1}W_{Y1}L_{X1}I_{X1}} \right] + \Delta V_{THY1}(ss, ff) - \Delta V_{THX1}(ss, ff) \quad (1)$$

$$V_C = \Delta V_{thx0}(SS, FF) + V_{thx0} + \eta V_T \ln \left[\frac{L_{X0}I_{X0}}{N_{X0}W_{X0}I_0} \right] \quad (2)$$

As a result, V_{P1} exhibits a proportional-to-absolute-temperature (PTAT) + proportional to the faster-skewed process (PTFP) voltage characteristic, while V_C presents complementary-to-absolute-temperature (CTAT) + complementary to the faster-skewed process (CTFP) behavior. The simulated temperature characteristics of V_P and V_C under five process corners are also shown in Fig. 4 (a). By properly sizing the transistors, both temperature variations and process-induced threshold voltage shifts can be effectively compensated, thereby generating a stable reference voltage that is independent of temperature and process variations.

In the proposed design, a four-stage cascaded PTAT + PTFP generator is employed to achieve a compensated reference voltage of 1.15 V. Moreover, this voltage reference avoids the use of large resistors commonly required in ULP circuits, which significantly reduces the area overhead.

Fig. 4 (b) illustrates the simplified structure of the adaptive FVF LDO used in our system. The LDO provides fast transient response and excellent stability. When the power switch is turned off, the entire system maintains nW-level power consumption, enhancing the sensitivity of the PMU-integrated rectifier. In addition, the FVF architecture eliminates the need for large capacitors and resistors, enabling compact on-chip integration without compromising performance.

III. RESULTS

The proposed surge-based RFEH power supplying system was designed and simulated using a standard 40-nm CMOS process, occupying a core area of 427×206 μm^2 , as shown in Fig. 5. Under the predefined operating voltage of 1.5 V and no-load conditions, the CMOS voltage reference and the LDO consume 510 nW and 321 nW, respectively, resulting in a total power consumption of 831 nW. Therefore, the system can rapidly accumulate energy under this light-load condition.

Fig. 6 (a) shows the output reference voltage of the CMOS voltage reference over a temperature range from $-40^\circ C$ to $80^\circ C$. As shown in Fig. 6 (b), the results of 200 Monte Carlo simulations demonstrate effective compensation for both process and mismatch variations, achieving a σ/μ ratio of 0.674%, which highlights the excellent stability of the proposed design. In addition, the LDO achieves a power supply rejection ratio (PSRR) of -34 dB at 1 kHz, effectively suppressing disturbances caused by rectifier output fluctuations

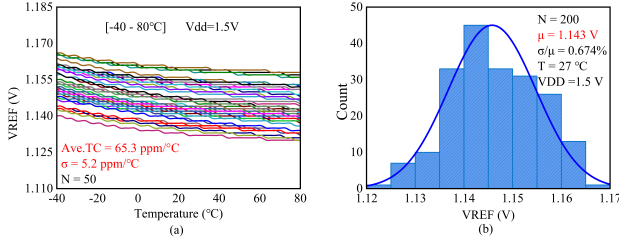


Fig. 6. (a) Post-layout simulated VREF versus temperature at VDD = 1.5 V. (b) Distribution of post-layout simulated VREF.

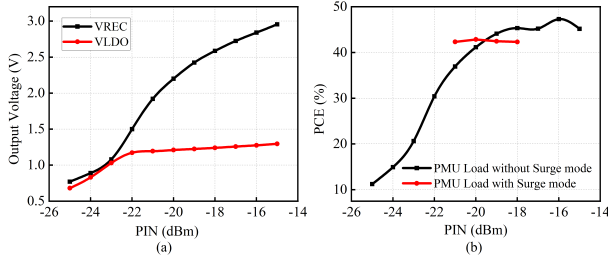


Fig. 7. (a) Post-layout simulated results of rectifier and LDO output voltages versus input RF power. (b) Comparison of post-layout simulated PCE under PMU load: with surge mode vs. without surge mode.

when the system operates in the surge mode, thereby providing a stable supply voltage for functional circuits during activation.

Fig. 7 (a) illustrates the relationship between the output voltage of the RFEH front-end and the input power when the surge-mode operation is not yet activated, as well as the LDO output voltage under different input conditions. It can be observed that when the PMU remains active, the rectifier output voltage reaches 1.5 V once the input power exceeds -22 dBm, while the LDO provides a stable 1.2 V supply voltage. This indicates that when the input power is higher than -22 dBm, the system can operate in the predefined surge mode to power the subsequent circuits. Moreover, although the LDO operates between 1.5 V and 1.6 V during the actual surge mode, its wide supply range and low line regulation further enhance the overall system stability.

Fig. 7 (b) separately shows the PCE under the non-surge operating mode compared with that in the surge mode. Specifically, the system achieves a peak PCE of 42.25% at an input power of -20 dBm. Even as the input power increases to -18 dBm, the surge-mode limits the output voltage to 1.6V, suppressing leakage and maintaining the PCE above 40% across the entire measured input range.

Fig. 8 illustrates the transient characteristics of the proposed circuit operating in surge mode under an input power of -20 dBm. An 800 pF capacitor was used as the power storage element, and a 0.5 M Ω resistive load was employed to emulate the driven core circuit for the simulation. As shown, the circuit successfully performs the intended wave-like operation. The CMOS voltage reference first generates a stable reference voltage, and once sufficient energy is accumulated to reach the target voltage, the load is enabled. The LDO then quickly transitions to a heavy-load mode, providing a stable supply

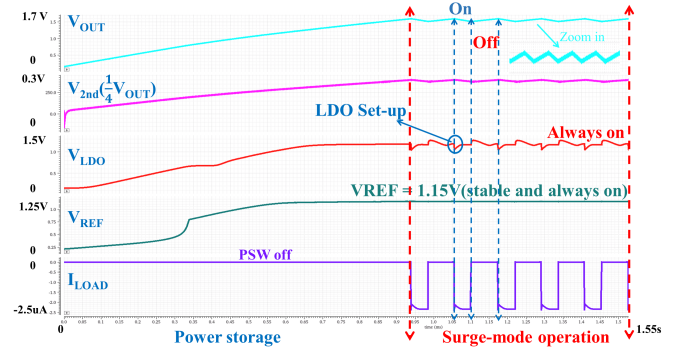


Fig. 8. Timing diagram of the proposed surge-mode RFEH power supplying system at -20dBm RF input.

TABLE I
PERFORMANCE SUMMARY OF THE RFEH FRONT-END

Parameters	This Work	TCAS-I'21 [2]	TMTT'23 [8]	TCAS-I'25 [9]
Technology	40 nm	130 nm	28 nm	55 nm
Frequency	915 MHz	915 MHz	970 MHz	915 MHz
Chip Area (mm^2)	0.088	0.029	0.025	0.027
No. of Stages	8 + PMU	10	108	3 + LFCE
Sensitivity (Vout @ 1.5V)	-22 dBm @ PMU (Load Independent)	-20.5 dBm @ 1 M Ω	-36.5 dBm @ ∞ (Vout = 1)	-25.7 dBm @ LFCE
Peak PCE & R_L	42.25% (surge-mode) @ -20 dBm & PMU	42.4% @ -16 dBm & 0.45 M Ω	1.22% @ -13 dBm & NA	31.3% @ -25 dBm & LFCE

* Surge-Mode Advantage: Driving capability is decoupled from input power. Input power only determines the operation frequency, enabling the system to drive heavy loads once PMU activated (> -22 dBm).

voltage to the load, while the rectifier output voltage gradually decreases until the load is turned off, initiating the next charging phase. Under different input power levels, the energy accumulation rate varies, thereby enabling an adaptive, self-regulated wave-like operation. It is worth noting that, in practical systems, the storage capacitor must be sized based on the actual load and operating duration. This ensures that the voltage decay time to the minimum threshold exceeds the sum of the LDO settling time and the system's active period.

Table I presents a summary of the proposed system and compares it with state-of-the-art works. This work presents a fully integrated design of a periodically activated self-powered system based on RFEH, which achieves high sensitivity and stable, efficient operation over a wide input range.

IV. CONCLUSION

A surge-mode RFEH front-end has been demonstrated to achieve high sensitivity and stable performance under ultra-low input power. By integrating an nW-level voltage monitor and an always-active ULP PMU, the proposed system enables adaptive intermittent operation and maintains efficiency across varying input conditions. Fabricated in 40-nm CMOS, the design achieves -22 dBm sensitivity and 42.25% peak PCE at 915 MHz, demonstrating its strong potential for next-generation self-powered IoT and biomedical devices.

ACKNOWLEDGMENT

This work was supported by the Brain Science and Brain-like Intelligence Technology-National Science and Technology Major Project (Grant No.2022ZD0208805), "Pioneer" and "Leading Goose" R&D Program of Zhejiang (2024C03002).

REFERENCES

- [1] Changgui Yang, Yunshan Zhang, Ziyi Chang, Zhuhao Li, Huan Gao, Yuxuan Luo, Kedi Xu, Gang Pan, and Bo Zhao. Neural dielet: A 0.4 mm³ battery-less crystal-less neural-recording system on die achieving 1.6 cm backscatter range with 2 mm × 2 mm on-chip antenna. *IEEE Transactions on Biomedical Circuits and Systems*, 17(1):54–66, 2022.
- [2] S Nagaveni, Praveen Hunasigidad, Deepali Pathak, and Ashudeb Dutta. On-chip configurable rf energy harvester for biomedical implantable devices. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 2024.
- [3] Seyed Mohammad Noghabaei, Rafael L Radin, Yvon Savaria, and Mohamad Sawan. A high-sensitivity wide input-power-range ultra-low-power rf energy harvester for iot applications. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 69(1):440–451, 2021.
- [4] Haoyu Jiang, Zihan Wu, and Hao Min. Co-design of an auxiliary self-bias module and a cmos rectifier for rf energy harvesting. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 71(3):1481–1485, 2023.
- [5] Xing Liu, Jinbo Chen, Hui Wu, Wenjun Zou, Jie Yang, and Mohamad Sawan. An on-chip reconfigurable front-end for ultra-low-power rf energy harvesting. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 2025.
- [6] Mohamed A Abouzied, Krishnan Ravichandran, and Edgar Sánchez-Sinencio. A fully integrated reconfigurable self-startup rf energy-harvesting system with storage capability. *IEEE Journal of Solid-State Circuits*, 52(3):704–719, 2016.
- [7] Yang Zhao and Yong Lian. Event-driven circuits and systems: A promising low power technique for intelligent sensors in aiot era. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 69(7):3122–3128, 2022.
- [8] Yoomi Park and Sangjin Byun. Analysis and design of a 970-mhz, 108-stage cmos ambient rf energy harvester with- 36.5-dbm input power sensitivity. *IEEE Transactions on Microwave Theory and Techniques*, 72(6):3829–3840, 2023.
- [9] Haoyu Jiang, Zihan Wu, and Hao Min. A-31 dbm sensitivity high-efficiency rf energy harvesting system with burst charging mode for iot applications. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 2025.
- [10] Wenjun Gong, Binbin Chen, Liangjian Lyu, and C-J Richard Shi. A 240-mv 33.8- $\mu\text{V}/^\circ\text{C}$ 1.5-nw voltage detector for energy harvesting. In *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–5. IEEE, 2025.
- [11] Jing Wang, Jun He, Man-Kay Law, Xinzhe Wang, Futian Liang, and Lin Cheng. A 76.9 ppm/k nano-watt pvt-insensitive cmos voltage reference operating from 4 to 300 k for integrated cryogenic quantum interface. *IEEE Journal of Solid-State Circuits*, 2025.
- [12] Jing Wang, Xuan Sun, and Lin Cheng. A picowatt cmos voltage reference operating at 0.5-v power supply with process and temperature compensation for low-power iot systems. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 70(4):1336–1340, 2022.
- [13] Yuanqing Huang, Yan Lu, Franco Maloberti, and Rui P Martins. Nano-ampere low-dropout regulator designs for iot devices. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(11):4017–4026, 2018.
- [14] Jung Sik Kim, Khurram Javed, Kyoung Hyun Min, and Jeongjin Roh. A 13.5-na quiescent current ldo with adaptive ultra-low-power mode for low-power iot applications. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 70(9):3278–3282, 2023.